
EvoXRB: From Transparent Synthetic Optimisation to a Preregistered Test of Disk–Corona/QPO Coupling in MAXI J1820+070

Software-methods paper and registered confirmatory protocol

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Publication status and claim boundary. The released software evaluated here is EvoXRB 0.2.0. “Version 3” denotes the third methodological generation and its preregistered implementation target; it is not yet a scientifically accepted package release. Results in this paper are synthetic validation results. The software includes the v0.3 analysis machinery, but no HXMT/NICER coupling result is reported because the pinned external XSPEC grid, immutable observational products, full blinded calibration, and observational campaign have not yet passed their non-waivable gates.

Article type	Software methods, validation study, and registered confirmatory protocol
Current release	EvoXRB 0.2.0
Canonical bundle	Smoke profile, seed 1,820,070, configuration hash d17357d31d720259
Primary target	MAXI J1820+070, with HXMT primary, NICER sensitivity, and MAXI cross-check
Code state	Analysis implementation complete; external scientific acceptance pending

Abstract

Black-hole X-ray binary inference couples detector response, non-Gaussian count statistics, multimodal optimisation, posterior uncertainty, and red-noise timing. EvoXRB was developed through three methodological generations to make that chain inspectable. Version 0.1.0 established a deterministic synthetic forward-folding pipeline with Poisson simulation, Cash-statistic fitting, a bounded real-coded genetic algorithm, local optimisation, posterior sampling, and synthetic timing recovery. Version 0.2.0 added observable-generation callbacks, checkpoint/replay products, explicit provenance, recovery studies, and strict separation between synthetic likelihood inputs and a real MAXI/GSC contextual spectrum. In the canonical smoke bundle (configuration hash d17357d31d720259), a deliberately short raw genetic search had 100% failure under a $\Delta C > 1$ criterion, whereas local L-BFGS-B polishing reduced the failure rate to zero and the mean gap from the multi-start optimum to 1.27×10^{-8} . At 512 s, correct-model root-mean-square errors were 0.00206 keV in disk temperature, 283 in disk normalisation, 0.0129 in photon index, and 0.00582 in continuum normalisation. A misspecified cutoff-truth/power-law-fit control reached the same numerical optimum yet retained material physical bias, demonstrating that optimisation success is not model adequacy. A synthetic 5 Hz type-C-like QPO was recovered at 5.0277 ± 0.0061 Hz, while a low-quality 0.05 Hz candidate and a no-QPO epoch remained non-detections.

The proposed v0.3.0 study is therefore specified prospectively: calibrated OGIP likelihoods, a pinned XSPEC 13.0.0 disk–corona–reflection grid, joint spectral/timing epoch posteriors, posterior-recycled population inference, four coupling hypotheses, PSIS-LOO with mandatory exact refits, contiguous-block validation, and blinded simulation-based acceptance. This paper reports validated software evidence and a falsifiable analysis protocol; it makes no observational claim that luminosity, photon index, optical depth, or QPO frequency are causally coupled.

Keywords: black-hole X-ray binaries; MAXI J1820+070; forward folding; genetic algorithms; Bayesian workflow; quasi-periodic oscillations; reproducible research; preregistration.

1 Introduction

Accreting stellar-mass black holes provide rapidly evolving laboratories for disk, corona, jet, and relativistic variability physics. Their outbursts are commonly organized into hard, intermediate, and soft states, while their power spectra exhibit broad-band noise and quasi-periodic oscillations (QPOs) whose centroid frequencies and quality factors change with spectral state [1, 2]. Low-frequency type-C QPOs are particularly informative because their evolution has been associated with changes in the inner accretion flow and can be interpreted, under specific geometric assumptions, through Lense–Thirring precession [3]. Those physical interpretations remain model dependent: an observed correlation is not by itself evidence of a unique mechanism.

MAXI J1820+070 is an unusually strong test case. It was monitored across its 2018 outburst by MAXI, Insight-HXMT, NICER, Swift, and multi-wavelength facilities [4–8]. Reverberation measurements suggested contraction of the corona during the hard state [9]. Independent dynamical and continuum-fitting work constrains the distance to 2.96 ± 0.33 kpc, gives a mass near $8.48^{+0.79}_{-0.72} M_{\odot}$ if the 63 ± 3 degree radio-jet orientation is adopted, and favors a low spin $a_* = 0.14 \pm 0.09$ [10–12]. A recent HXMT/NICER analysis reported a positive association between type-C QPO frequency and photon index Γ , together with a negative association with coronal optical depth τ [13]. Reproducing and stress-testing those associations requires more than refitting a plotted trend: detector responses, background treatment, spectral covariance, QPO uncertainty, non-detections, shared distance, model variants, and time dependence must all enter the inferential target.

EvoXRB began as an educational synthetic X-ray binary optimizer and has matured into an auditable scientific workflow. The central contribution of this paper is not an observational correlation estimate. It is the disciplined separation of three things that are often conflated:

1. **validated software behavior**, for which executable tests and quantitative artifacts exist;
2. **synthetic scientific evidence**, which can establish computational recovery but not an astrophysical fact; and

3. a **registered observational protocol**, whose claims remain locked until external data and calibration gates pass. This structure follows reproducible-computing guidance that data transformations, seeds, environments, and intermediate products should be inspectable [14–16]. Figure 1 summarizes the versioned progression and the scientific release boundary.

EvoXRB methodological progression and the claim boundary

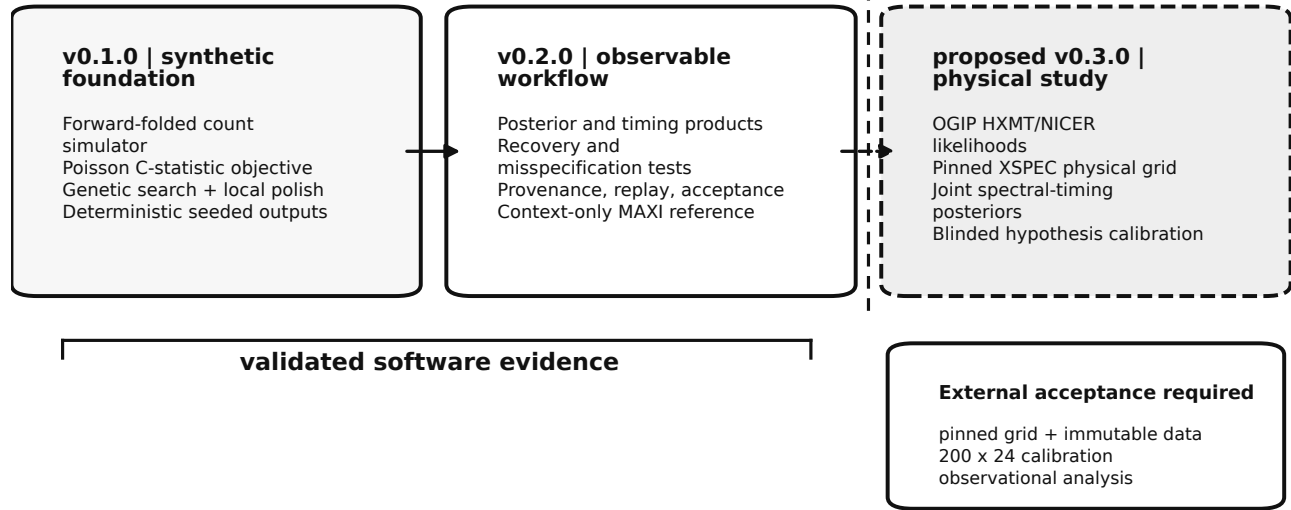


Figure 1: Methodological progression of EvoXRB. Solid boxes summarize validated released behavior; the dashed v0.3 box is implemented as a registered analysis path but remains outside the scientific claim boundary. “Validated” refers to software acceptance and synthetic recovery, not observational confirmation.

2 Scientific prerequisites and theoretical basis

The analysis begins with physical parameters but ends with integer counts and sampled light curves. The intervening steps are part of the model, not bookkeeping. At a high level, the spectral inference chain is

$$\theta \xrightarrow{\text{radiative physics}} S(E; \theta) \xrightarrow{\text{instrument}} \{\mu_i(\theta)\} \xrightarrow{\text{counting noise}} \{y_i\} \xrightarrow{\text{likelihood}} p(\theta | y). \quad (1)$$

Here θ denotes the physical parameters, $S(E; \theta)$ is the photon spectrum arriving at the telescope, μ_i is the expected number of counts in detector channel i , and y_i is the observed integer count. This section derives each arrow and explains the assumptions behind it.

2.1 Forward-folded X-ray spectroscopy

2.1.1 From incident photons to detector channels

Let $S(E; \theta)$ have units of photons $\text{cm}^{-2} \text{s}^{-1} \text{keV}^{-1}$. In a small energy interval dE and time interval dt , the expected number of photons intercepted by a collecting area $A(E)$ is

$$dN_{\text{inc}} = S(E; \theta) A(E) dE dt. \quad (2)$$

This is simply “rate density $\times$ bandwidth $\times$ time.” The effective area is smaller than the geometric mirror area because it includes losses from reflectivity, filters, detector efficiency, and other throughput terms.

The detector does not usually report the true photon energy. If $R_i(E)$ is the conditional probability that a detected photon of true energy E is assigned to channel i , then the expected contribution to that channel is $R_i(E) dN_{\text{inc}}$. Independent energy intervals add linearly, so an exposure t gives

$$\mu_i(\theta) = t \int R_i(E) A(E) S(E; \theta) dE + b_i. \quad (3)$$

The additive term b_i is the expected background count in the source region. In a discretized response with true-energy bin j and width ΔE_j , the same equation is the matrix multiplication

$$\mu_i = t \sum_j R_{ij} A_j S_j \Delta E_j + b_i. \quad (4)$$

The redistribution matrix file (RMF) supplies R_{ij} ; the auxiliary response file (ARF) supplies A_j ; an RSP may contain their product. This is why a model curve cannot be compared directly with a channel-count spectrum: the curve must first be blurred and weighted by the particular instrument [17–19].

Equation 4 also explains why “unfolding” is not a harmless inverse operation. Neighboring true-energy bins can produce very similar channel patterns, so the response matrix can be nearly singular. Forward folding instead proposes a physically constrained $S(E; \theta)$, predicts the counts, and compares those predictions with the data. The v0.3 path therefore propagates exposure, BACKSCAL, AREASCAL, quality, and grouping metadata and blocks ambiguous RSP/RMF area combinations.

2.1.2 From photon counting to the Cash statistic

Photon arrivals are discrete. Divide an exposure into a very large number M of tiny intervals, each with a very small probability p of containing a photon. The count is initially binomial. Taking the limit $M \rightarrow \infty$, $p \rightarrow 0$, while keeping $Mp = \mu_i$ fixed gives the Poisson distribution

$$p(y_i | \mu_i) = \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!}, \quad y_i = 0, 1, 2, \dots \quad (5)$$

Conditional on the spectral model, counts in disjoint channels are treated as independent. Their joint likelihood is therefore a product. Taking its logarithm converts the product into a sum:

$$\ln \mathcal{L}(\theta) = \sum_i [y_i \ln \mu_i(\theta) - \mu_i(\theta) - \ln(y_i!)] . \quad (6)$$

Maximum likelihood chooses the parameters that make the observed counts most probable. Terms containing only y_i do not change when θ changes, but retaining an appropriate data-only term makes the statistic interpretable as a likelihood ratio. The saturated model assigns one free mean to every channel and fits the data exactly, $\mu_i = y_i$. Twice the negative log likelihood ratio between the proposed and saturated models is the Cash deviance [20]:

$$C(\theta) = -2 \ln \left(\frac{\mathcal{L}(\theta)}{\mathcal{L}_{\text{sat}}} \right) = 2 \sum_i \left[\mu_i - y_i + y_i \ln \left(\frac{y_i}{\mu_i} \right) \right]. \quad (7)$$

For $y_i = 0$, the limit $y_i \ln(y_i/\mu_i) \rightarrow 0$ is used. Thus empty channels remain valid information: a model predicting many counts where none were observed is penalized. Near the optimum and at high counts, a Taylor expansion gives

$$C_i \simeq \frac{(y_i - \mu_i)^2}{\mu_i}, \quad (8)$$

which is the familiar squared-residual form. The Poisson derivation is nevertheless preferable because it remains valid without assuming that every channel contains enough counts to be approximately Gaussian.

Background data require one more layer. Let m_i be the predicted source count, β_i the mean count in a measured background region, α_i the exposure/area scaling into the source region, and z_i the observed background-region count. The generative statement is

$$p(y_i, z_i | m_i, \beta_i) = \text{Pois}(y_i | m_i + \alpha_i \beta_i) \text{Pois}(z_i | \beta_i). \quad (9)$$

The difference of two Poisson variables is not Poisson, so background subtraction followed by Equation 7 would use the wrong sampling distribution. Profiling the nuisance β_i in Equation 9 leads to the W-statistic. If the background estimate instead has a Gaussian uncertainty, the second Poisson factor is replaced by a normal density, leading to PG-stat. The physical workflow implements these cases consistently with XSPEC [19]; the statistic is fixed by the data-generating process, not selected after inspecting the desired trend.

Figure 2 shows the educational v0.2 response. Its effective area and Gaussian redistribution are deliberately transparent and NICER-inspired, not a calibrated NICER response. That distinction is printed on every generated artifact.

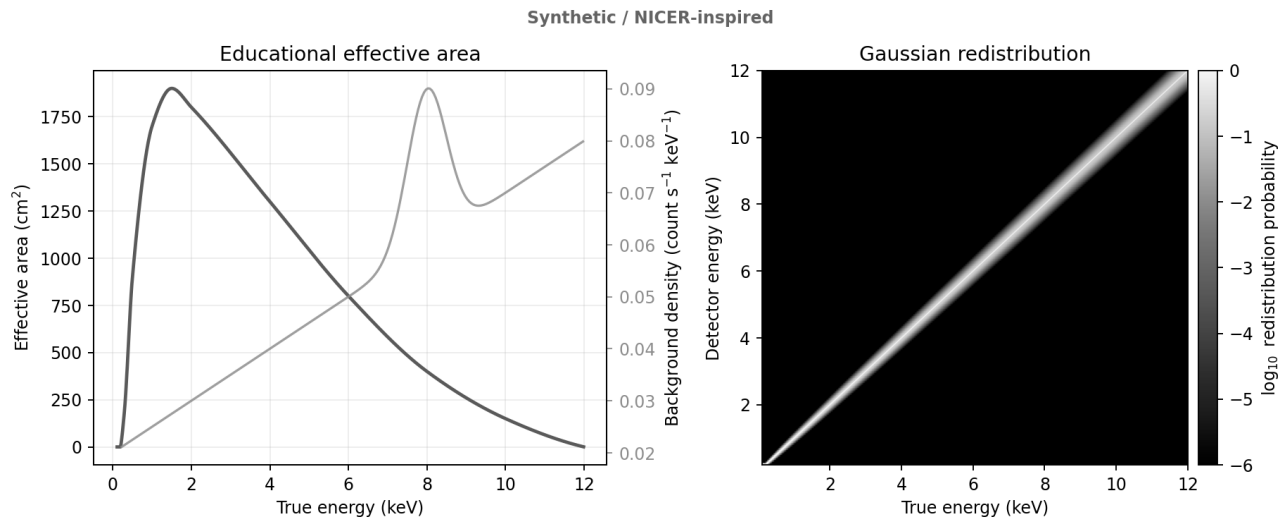


Figure 2: Educational response used for the synthetic validation. The left panel gives the assumed effective area and background density; the right panel gives the redistribution kernel. These curves support method testing only and are not substituted for calibrated instrument files in v0.3.

2.2 Disk, corona, absorption, and reflection

2.2.1 Thermal emission from a geometrically thin disk

A surface in local thermal equilibrium emits a Planck spectrum. Per unit photon energy, the spectral radiance is

$$B_E(T) = \frac{2E^3}{h^3 c^2} \frac{1}{\exp(E/k_B T) - 1}, \quad (10)$$

where h is Planck's constant, c is the speed of light, and k_B is Boltzmann's constant [21]. Dividing B_E by E gives the corresponding photon-number spectrum. A disk is not at one temperature. In a Newtonian thin disk, conservation of gravitational energy and angular momentum gives the radiative flux from one face at radius r ,

$$F(r) = \frac{3GM\dot{M}}{8\pi r^3} \left[1 - \left(\frac{R_{\text{in}}}{r} \right)^{1/2} \right] = \sigma_{\text{SB}} T_{\text{eff}}^4(r), \quad (11)$$

for central mass M , accretion rate $\dot{M}$, and inner radius R_{in} [22]. Away from the inner boundary, the bracket is slowly varying, so $T_{\text{eff}}(r) \propto r^{-3/4}$. The observed disk spectrum is the sum of blackbodies from annuli with area $2\pi r dr$:

$$L_E \propto \int_{R_{\text{in}}}^{R_{\text{out}}} 2\pi r B_E[T_{\text{col}}(r)] dr. \quad (12)$$

This multicolor sum is broader than a single-temperature blackbody and is the basis of `diskbb` [23]. Its fitted normalization is approximately

$$N_{\text{disk}} = \left(\frac{R_{\text{in,app}}/\text{km}}{D/(10 \text{ kpc})} \right)^2 \cos i, \quad (13)$$

where $R_{\text{in,app}}$ is an apparent inner radius, D is distance, and i is inclination. Boundary stress, color correction, relativistic transfer, and calibration alter the conversion from $R_{\text{in,app}}$ to a physical radius; Equation 13 should therefore not be interpreted geometrically without those corrections.

2.2.2 Photoelectric absorption

Interstellar gas removes photons from the line of sight. If I_E is the specific intensity and $n_H \sigma_{\text{eff}}(E)$ is the absorption probability per unit path length, radiative transfer without in-scattering gives

$$\frac{dI_E}{ds} = -n_H(s) \sigma_{\text{eff}}(E) I_E. \quad (14)$$

Separating variables and integrating produces the Beer–Lambert form

$$I_E^{\text{obs}} = I_E^{\text{int}} \exp[-N_H \sigma_{\text{eff}}(E)], \quad N_H = \int n_H(s) ds. \quad (15)$$

The effective cross section sums the contributions of the elements and their adopted abundances. Because photoelectric cross sections rise steeply toward low energy, N_H is most strongly constrained by the soft spectrum. `TBabs` implements this energy-dependent transmission with modern interstellar abundances and cross sections [24].

2.2.3 Thermal Comptonization in a hot corona

Disk photons can scatter from hot electrons above the disk. Define the dimensionless electron temperature $\Theta_e = kT_e/(m_e c^2)$. For non-relativistic thermal electrons and seed photons well below the electron energy, the mean fractional energy gain per scattering is approximately $4\Theta_e$. A photon scatters roughly τ times when the Thomson optical depth $\tau \ll 1$, and roughly τ^2 times when it performs a random walk through an optically thick cloud. Their product gives the Compton parameter

$$y_C \simeq 4\Theta_e \max(\tau, \tau^2), \quad (16)$$

which measures the total fractional energy change before escape [21, 25]. When Comptonization is unsaturated, repeated small gains transform a seed spectrum into an approximate power law between the seed-photon energies and a rollover set by kT_e . A useful diffusion-limit intuition is

$$\alpha \simeq -\frac{3}{2} + \sqrt{\frac{9}{4} + \frac{4}{y_C}}, \quad \Gamma = \alpha + 1, \quad (17)$$

although the exact mapping depends on geometry, seed-photon distribution, and relativistic corrections. `ThComp` solves a more accurate transfer problem and parameterizes the continuum by Γ (or τ), kT_e , and covering fraction [26]. Equation 16 explains the important covariance: different combinations of temperature and optical depth can produce similar slopes.

2.2.4 Reflection and relativistic transfer

Some coronal photons illuminate optically thick material rather than escaping directly. Photoelectric absorption followed by fluorescent emission produces features such as Fe $K\alpha$, while repeated Compton scattering produces a broad high-energy hump. `xillver` calculates this rest-frame reprocessed spectrum for an illuminated slab [27]. Close to a black hole, Doppler shifts, gravitational redshift, and light bending redistribute the emitted energies. Schematically, the observed relativistic reflection is a convolution

$$F_{\text{refl}}^{\text{obs}}(E) = \int K(E, E'; a_*, i, R_{\text{in}}, q) F_{\text{refl}}^{\text{rest}}(E') dE', \quad (18)$$

where the transfer kernel K depends on spin a_* , inclination i , inner radius, and illumination profile q . `relxillCp` combines this relativistic transfer with an ionized reflection calculation [28]; a separate `xillverCp` component represents more distant, comparatively narrow reflection.

The released simulator uses an explicitly educational absorption factor multiplied by a disk-like component and either a power law or a cutoff surrogate. It is useful for testing identifiability, parameter mappings, and optimizers, but its parameters must not be reported as calibrated measurements. The proposed physical study instead fixes the primary expression before examining the target association:

$$\text{TBabs} \times [\text{ThComp}(\text{diskbb}) + \text{relxillCp}_{\text{refl}} + \text{xillverCp}_{\text{refl}}]. \quad (19)$$

The multiplication means that the interstellar transmission affects every emitted component. The terms inside brackets are additive photon sources: Comptonized disk photons, relativistically blurred reflection, and distant reflection. Both reflection components are restricted to reflection-only mode so that the direct coronal continuum is counted once rather than once inside each reflector. Spin ($a_* = 0.14$), inclination (63°), emissivity index (3), redshift (0), iron abundance (solar), and baseline column density [$N_{\text{H}} = 0.15 \times 10^{22} \text{ cm}^{-2}$] are fixed from the registered configuration and prior work [10–13]. Sensitivity variants remove reflection and allow a shared free N_{H} .

Direct repeated XSPEC calls would make a large calibration campaign expensive and difficult to audit. The protocol therefore requires a versioned grid generated with XSPEC 13.0.0 over 0.01–1000 keV, using pinned `ThComp` and `relxill/xillver v2.3` components. Interpolation is allowed only inside the grid. Off-grid direct-XSPEC comparisons are a release gate, not an optional diagnostic.

2.3 Power spectra and QPO evidence

2.3.1 From a light curve to a power spectrum

Let x_n be a uniformly sampled count-rate light curve with sample spacing Δt , sample mean $\bar{x}$, and N points. The discrete Fourier transform of the mean-subtracted light curve is

$$X_k = \sum_{n=0}^{N-1} (x_n - \bar{x}) \exp\left(-\frac{2\pi i k n}{N}\right), \quad \nu_k = \frac{k}{N\Delta t}. \quad (20)$$

The exponential is a rotating unit vector. Multiplication by it asks how strongly the light curve aligns with a sinusoid at frequency ν_k . Squaring the Fourier amplitude discards phase and yields a periodogram. One common fractional-rms normalization is

$$I(\nu_k) = \frac{2\Delta t}{N\bar{x}^2} |X_k|^2, \quad \sum_{k>0} I(\nu_k) \Delta \nu \simeq \frac{\text{Var}(x)}{\bar{x}^2}, \quad (21)$$

where $\Delta \nu = 1/(N\Delta t)$. The area under the positive-frequency power spectrum is therefore the fractional variance. Its square root is the fractional root-mean-square variability, which gives a direct physical meaning to the normalization.

2.3.2 Why Lorentzians describe broad noise and QPOs

A perfectly coherent sinusoid observed forever would give a delta-function peak. Real oscillations lose phase coherence. If an oscillatory autocovariance decays exponentially,

$$C_j(\tau_{\text{lag}}) \propto e^{-2\pi\Delta_j|\tau_{\text{lag}}|} \cos(2\pi\nu_j\tau_{\text{lag}}), \quad (22)$$

its Fourier transform is a pair of Lorentzian peaks at $\pm\nu_j$. Around the positive-frequency peak, the component can be written

$$L_j(\nu) = \frac{r_j^2\Delta_j}{\pi[(\nu - \nu_j)^2 + \Delta_j^2]}, \quad \int_{-\infty}^{\infty} L_j(\nu) d\nu = r_j^2. \quad (23)$$

The implementation uses the corresponding one-sided normalization, including the negative-frequency mirror in the integrated fractional variance. At $\nu = \nu_j \pm \Delta_j$, the height is half the peak height, so the full width at half maximum is $\text{FWHM} = 2\Delta_j$. The quality factor is therefore

$$Q_j = \frac{\nu_j}{\text{FWHM}} = \frac{\nu_j}{2\Delta_j}. \quad (24)$$

For example, a peak at 5 Hz with FWHM 0.6 Hz has $Q \simeq 8.3$: it persists for several cycles but is not perfectly coherent. Broad, zero-centered Lorentzians describe aperiodic variability; narrower, non-zero-centered Lorentzians describe QPO candidates.

2.3.3 The Whittle likelihood

The periodogram is not a smooth measurement with a small Gaussian error bar. Even for a fixed underlying power spectral density $P(\nu)$, individual powers fluctuate strongly. For a sufficiently long, stationary time series, the real and imaginary Fourier coefficients at distinct Fourier frequencies are approximately independent Gaussian variables. Their squared magnitude is then exponentially distributed:

$$p[I_k | P_k(\phi)] = \frac{1}{P_k(\phi)} \exp\left[-\frac{I_k}{P_k(\phi)}\right], \quad (25)$$

where $I_k = I(\nu_k)$ and $P_k = P(\nu_k; \phi)$ is the model power. Multiplying over frequencies and taking minus twice the log gives the Whittle statistic

$$-2 \ln \mathcal{L}_W(\phi) = 2 \sum_k \left[\ln P_k(\phi) + \frac{I_k}{P_k(\phi)} \right] + \text{constant}. \quad (26)$$

This is the frequency-domain analogue of using the correct Poisson likelihood for counts: the assumed noise distribution determines the objective function [29, 30]. The approximation becomes imperfect for short segments, strong leakage, gaps, or non-stationarity, which is why segment definition and validation matter.

The timing likelihood is defined on 512 s segments over 0.01–64 Hz. Its model combines white noise, up to four broad Lorentzians, a type-C component, allowed harmonic/subharmonic components, and a separate type-B component. Red-noise model uncertainty is material to QPO searches [31]; synthetic light curves therefore use power-law-noise simulation rather than adding sinusoidal peaks to white noise [32]. Candidate type-C components must satisfy the registered $2 \leq Q \leq 30$ interval and evidence requirements. Non-detections are retained and enter the population model through a QPO/noise evidence mixture.

2.4 Computational prerequisites

The likelihood connects parameters to data, but it is not yet an uncertainty statement. Bayes' theorem combines it with a prior density,

$$p(\boldsymbol{\theta} | \mathbf{y}) = \frac{p(\mathbf{y} | \boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{y} | \boldsymbol{\theta}')p(\boldsymbol{\theta}') d\boldsymbol{\theta}'}, \quad (27)$$

to produce the posterior distribution. An optimizer identifies a high-likelihood or high-posterior point; it does not describe the volume, asymmetry, or degeneracy of the surrounding uncertainty. Sampling methods are used because many combinations, such as (kT_e, τ) or (R_{in}, D, i) , can predict similar observables.

A posterior predictive check closes the generative loop:

$$p(\mathbf{y}^{\text{rep}} | \mathbf{y}) = \int p(\mathbf{y}^{\text{rep}} | \boldsymbol{\theta})p(\boldsymbol{\theta} | \mathbf{y}) d\boldsymbol{\theta}. \quad (28)$$

It asks whether replicated counts or powers generated from posterior draws resemble the data in features that matter scientifically. A precise posterior under an inadequate model can fail this check, just as the misspecification experiment later in this paper reaches a numerical optimum while retaining physical bias.

Table 1 states the minimum prerequisites for interpreting each layer. These are inferential prerequisites, not merely software dependencies. Each row protects one link in Equation 1: calibrated response information protects the instrument mapping, likelihood choice protects the noise model, posterior diagnostics protect uncertainty propagation, and time/population controls protect the final scientific comparison.

Table 1: Prerequisites and failure modes addressed by the workflow.

Layer	Required object	Failure prevented
Spectral data	Source/background PHA, response, area, scaling, quality, grouping	Fitting display-space fluxes or mis-scaled backgrounds
Physical model	Pinned component versions, bounded grid, direct-model validation	Extrapolation and silent component-version drift
Optimization	Transformed bounded parameters, multi-start reference, local polish	Treating a stochastic search endpoint as the optimum
Posterior	Convergence diagnostics, effective sample sizes, posterior predictive checks	Reporting unstable intervals from unrepresentative chains
Timing	Segment definition, normalization, red-noise continuum, QPO/noise evidence	Selecting only visually prominent or significant peaks
Population	Shared distance, epoch posterior uncertainty, time blocks, phase offsets	Regression on point estimates and pseudoreplication
Reproducibility	Hashes, seeds, environment, immutable manifests, acceptance report	Results that cannot be reconstructed or distinguished from context data

3 Versioned methodological development

The project history records a progression from a deterministic simulator to a release-gated study design. Table 2 identifies what each generation can and cannot support. This explicit scope is essential because “more realistic code” does not retroactively convert synthetic results into observations.

Table 2: Scientific capability by methodological generation.

Generation	Primary contribution	Validated use	Excluded claim
v0.1.0	Deterministic response folding, Poisson simulation, bounded genetic optimization, local refinement, posterior and timing prototypes	Algorithmic behavior on known synthetic truth	Calibrated instrument or source parameters
v0.2.0	Generation callbacks, replay, checkpointing, recovery/misspecification controls, provenance, acceptance bundle, contextual MAXI reference	Reproducible synthetic method validation and educational inspection	Any likelihood contribution from MAXI/GSC or any physical correlation in J1820
Proposed v0.3.0	OGIP likelihoods, pinned physical grid, epoch spectral/timing inference, population model comparison, blinded calibration and release gates	Registered protocol implementation; machine-readable blocked state	Observational result before external assets and every gate pass

3.1 Generation 1: deterministic synthetic inference

The initial implementation separates parameter representation from physical coordinates. Parameters with large dynamic range, including disk normalization, are searched in transformed coordinates. A real-coded genetic algorithm uses bounded initialization, tournament selection, simulated binary crossover [33], mutation, elitism, and deterministic random-number streams. A Latin-hypercube multi-start local optimizer provides the reference solution [34]; the hybrid method polishes the genetic endpoint with L-BFGS-B [35, 36]. This design tests a practical scientific question: can the global search reach a basin from which a local method finds the best known count-space solution?

Posterior sampling is initialized only after optimization. The released path supports the affine-invariant ensemble method implemented by `emcee` [37]; the physical workflow additionally supports dynamic nested sampling and evidence accounting through `dynesty` [38]. Optimizer output and posterior output are not treated as interchangeable.

3.2 Generation 2: observable and auditable execution

Version 0.2.0 exposes generation-level callbacks, checkpoint/restart, animation and browser replay, parameter recovery summaries, timing products, and a provenance ledger. Every generated figure carries the label “Synthetic / NICER-inspired”. A real MAXI/GSC count-rate-density spectrum near the contextual E08 date is retained only as a cross-instrument visual reference. It is not numerically interchangeable with the simulated rate and is never included in Equation 7.

The acceptance bundle checks the required tables, spectra, figures, report, response, run summary, and posterior chain. Each run records a master seed, configuration hash, runtime, package version, Python/platform information, and dependency versions. A Windows-specific reproducibility audit also identified transient file-sharing failures during atomic checkpoint replacement. The checkpoint writer now retries bounded transient `PermissionError` failures while preserving atomic replacement; a focused regression test verifies recovery after two simulated sharing violations.

3.3 Generation 3: a release-gated physical association study

The v0.3 implementation includes sparse OGIP ingestion, calibrated statistic families, manifest hashing, physical-grid validation, epoch spectral/timing inference, luminosity propagation, coupling hypotheses, posterior recycling, predictive comparison, calibration, diagnostics, campaign orchestration, and blocked-result reporting. The version string intentionally remains 0.2.0. A semantic version change is justified only after the external grid, immutable HXMT/NICER assets, full calibration, and observational report pass in a clean environment.

4 Methods

4.1 Canonical synthetic case study

The canonical smoke profile is a bounded, fast acceptance case rather than a precision calibration. It uses master seed 1,820,070 and configuration hash d17357d31d720259. Twelve educational outburst epochs define evolving disk temperature T_{in} , disk-like normalization N_{disk} , photon index Γ , continuum normalization K , exposure, and optional QPO frequency. Three representative epochs (E01, E08, E10) are fully materialized for smoke-profile spectral/timing analysis. E08 has injected values $T_{\text{in}} = 0.65$ keV, $N_{\text{disk}} = 8000$, $\Gamma = 2.15$, and $K = 0.27$.

For each count spectrum, the simulator evaluates Equation 3, draws Poisson counts, and preserves the truth and seed. The fitting sequence is:

1. a multi-start SciPy reference fit;
2. a deliberately budget-limited genetic search (24 generations and 800 evaluations in the smoke profile);
3. L-BFGS-B polishing of the best genetic candidate; and
4. posterior sampling for E08.

The short genetic budget is a test of hybrid behavior, not the recommended stand-alone optimizer configuration.

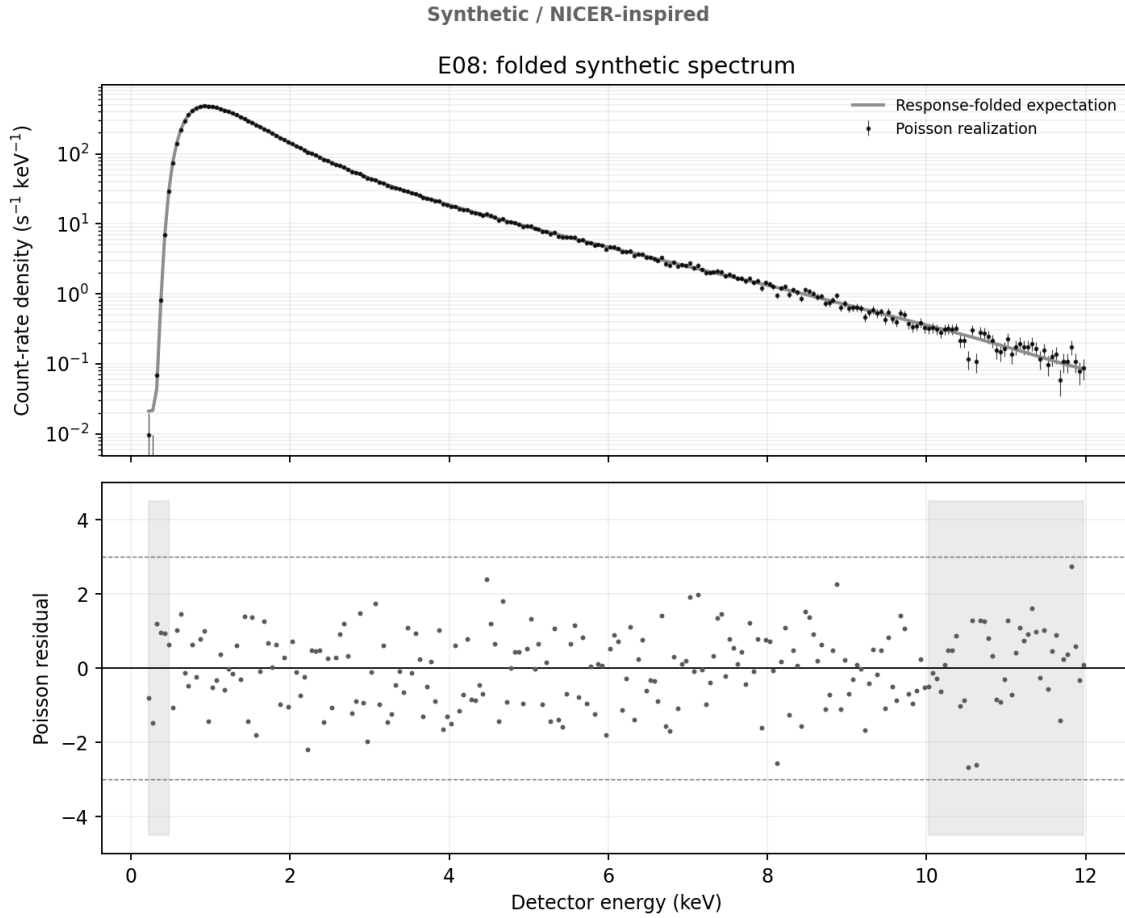


Figure 3: Response-folded E08 synthetic spectrum and Poisson residuals from the canonical smoke bundle. Grey bands mark excluded edge regions. The plotted reference date is contextual; the counts are simulated.

4.2 Recovery and model-misspecification controls

Recovery is evaluated against injected truth and against the best multi-start statistic. A run fails numerically when its ΔC exceeds 1 or violates a boundary/fit condition. The correct-model 512 s scenario has two realizations. A 4096 s high-signal scenario has one realization and is reported descriptively, not as a sampling distribution.

The misspecification control generates a spectrum with an educational 20 keV cutoff surrogate and fits it with the power-law model. It separates two hypotheses:

- H_{opt} : the optimizer reaches the best value within the fitted model;
- H_{model} : the fitted model recovers the generating physical parameters.

Passing H_{opt} cannot establish H_{model} .

4.3 Posterior and timing validation

The E08 posterior is sampled with `emcee` after hybrid optimization. The smoke acceptance checks sampler completion and finite ordered quantiles. It is not a substitute for the stricter v0.3 convergence requirements.

For timing, four 512 s segments are generated per selected epoch. E01 receives a 0.05 Hz, 30% fractional-rms injection; E08 receives a 5 Hz, 15% injection; and E10 has no injected QPO. The fit reports centroid, uncertainty,

width, Q , amplitude significance, bootstrap successes, and classification. The type-C-like label applies only to synthetic behavior and does not classify an observed J1820 epoch.

4.4 Focused full acceptance test

A separate opt-in acceptance test uses a 100 ks correct-model synthetic spectrum with $T_{\text{in}} = 0.7$ keV, $N_{\text{disk}} = 8000$, $\Gamma = 2.1$, $K = 0.25$, and $N_{\text{H}} = 0.15$. It compares a 20-start local reference against five independently seeded production-budget genetic runs, each locally polished. The test is preregistered to pass when at least four of five hybrids achieve $\Delta C \leq 1$, the best T_{in} and Γ errors are below 5%, and the best N_{disk} and K errors are below 10%.

4.5 Registered v0.3 observational protocol

4.5.1 Immutable data and objective inclusion

HXMT is the primary instrument, NICER is a separate sensitivity analysis, and MAXI is a cross-check. Every file must appear in an immutable manifest with byte size and SHA-256 hash. The inclusion/exclusion index is derived from data availability, instrument validity, and registered state/quality rules; epochs are never selected by QPO significance. The current manifests correctly mark the external HXMT/NICER collections and physical grid as unavailable.

4.5.2 Spectral grid and epoch fits

The grid spans the registered bounds in Table 3. It stores provenance for XSPEC and component versions, parameter axes, and energy bins. Interpolation occurs in log flux with analytic handling of linear parameters and no extrapolation. Each epoch is fitted independently under four variants: baseline reflection with fixed N_{H} ; no reflection; reflection with shared-free N_{H} ; and no reflection with shared-free N_{H} . Unabsorbed luminosity is integrated over 0.1–100 keV and shares a distance draw of 2.96 ± 0.33 kpc across epochs.

Table 3: Registered primary physical-grid bounds.

Parameter	Lower	Upper
Disk temperature T_{in} (keV)	0.05	2.0
Disk normalization N_{disk}	0.1	10^6
Electron temperature kT_e (keV)	5	300
Optical depth τ	0.05	10
Covering fraction	0	1
Ionization $\log \xi$	0	4.7
Inner radius (R_g)	1	1000

4.5.3 Timing fits and epoch posterior products

Timing models are fitted jointly across registered bands with a shared centroid where appropriate. Epoch-level products retain posterior draws of luminosity, Γ , $\log \tau$, QPO centroid, state/phase, and the log evidence for the QPO and noise alternatives. This product boundary allows the population analysis to recycle uncertainty without replacing a complex likelihood by a table of point estimates.

4.5.4 Population hypotheses

For epoch i , let z_L , z_Γ , and z_τ denote standardized $\log L_X$, Γ , and $\log \tau$. The core association model is

$$z_{\nu,i} = \alpha_{p[i]} + \beta_L z_{L,i} + \beta_\Gamma z_{\Gamma,i} + \beta_\tau z_{\tau,i} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_{\text{int}}^2), \quad (29)$$

where $\alpha_{p[i]}$ is a partially pooled phase offset. The four hypotheses in Table 4 differ only in registered slope constraints; β_L is retained as a common adjustment.

Table 4: Registered coupling hypotheses.

Hypothesis	Constraint	Interpretation
H_0	$\beta_\Gamma = \beta_\tau = 0$	QPO frequency is conditionally independent of the two coronal descriptors after luminosity and phase adjustment.
H_Γ	β_Γ free, $\beta_\tau = 0$	Photon index carries predictive association beyond luminosity/phase.
H_τ	$\beta_\Gamma = 0$, β_τ free	Optical depth carries predictive association beyond luminosity/phase.
$H_{\Gamma+\tau}$	β_Γ, β_τ free	Both descriptors contribute predictive information, subject to covariance and shrinkage.

Each epoch contributes through posterior recycling and a QPO/noise evidence mixture. Thus a non-detection changes the population likelihood without being discarded or assigned an arbitrary centroid. Shared distance is drawn once per population draw, preserving cross-epoch dependence.

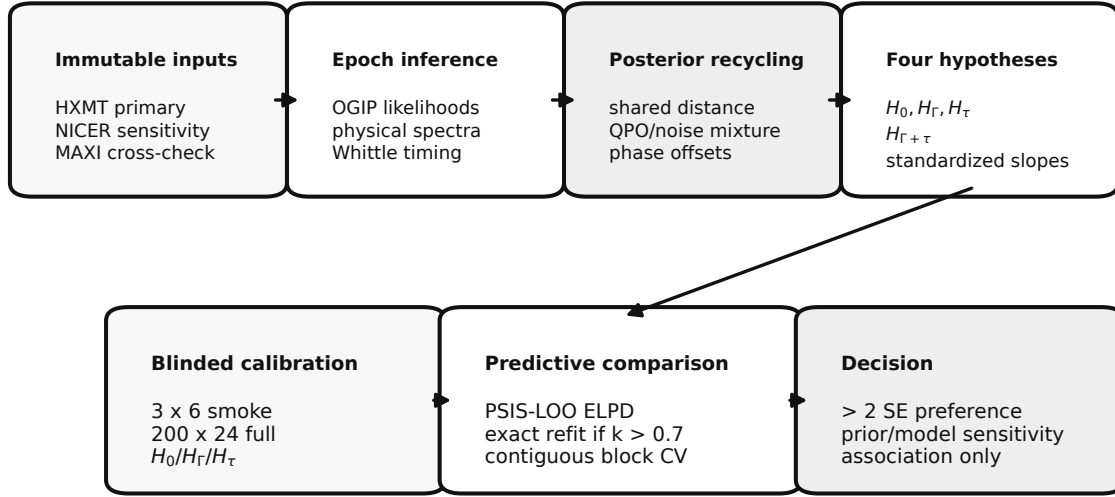
4.5.5 Predictive comparison and decision rule

The primary comparison is the expected log predictive density (ELPD) from Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) [39, 40]. Any observation with Pareto $\hat{k} > 0.7$ triggers an exact refit. Contiguous time-block cross-validation tests whether a preference generalizes beyond local temporal dependence. A model is preferred only when its ELPD improvement exceeds two standard errors and the direction is stable under:

- prior scales of 0.5, 1.0, and 2.0 times the primary slope scale;
- all four reflection/ N_H spectral variants;
- exact refits required by PSIS diagnostics; and
- the separate contiguous-block comparison.

Bayes factors are supporting summaries, not the primary selection rule. Rank-normalized $\hat{R} \leq 1.01$, bulk and tail effective sample sizes of at least 400, and nested-sampling evidence uncertainty below 0.1 log units are non-waivable [41, 42].

Preregistered v0.3 disk-corona/QPO association workflow



Unblinding and observational reporting are permitted only after all non-waivable gates pass.

Figure 4: Registered v0.3 analysis flow. Calibration, predictive comparison, and sensitivity analysis are part of the inferential design, not post hoc quality checks. Unblinding is permitted only after the complete blinded campaign finishes.

4.6 Blinded calibration and release gates

Calibration uses realistic instrument templates and shared spectral/timing latent variables. The smoke campaign comprises three realizations across six epochs. The full campaign comprises exactly 200 realizations for each primary data-generating hypothesis (H_0 , H_Γ , H_τ), with 24 epochs per realization and every spectral variant. Labels remain blinded until all jobs finish.

Table 5: Non-waivable v0.3 scientific acceptance gates.

Gate	Metric	Threshold
Physical grid	Folded flux error against direct XSPEC	Median $\leq 0.5\%$; 99th percentile $\leq 2\%$
Reference fit	Direct-grid statistic and parameter agreement	$ \Delta C \leq 1$; all parameters within 0.25σ
Interval recovery	90% interval coverage across full blinded calibration	0.85–0.95
False preference	Preferred non-null model under H_0	≤ 0.05
Power	Correct preference under H_Γ or H_τ	≥ 0.80
Direction	Correct recovered slope sign when non-null	≥ 0.90
Bias	Absolute standardized slope bias	≤ 0.15
Sampling	Rank-normalized $\hat{R}$; bulk/tail ESS	≤ 1.01 ; ≥ 400
Evidence	Nested-sampling log-evidence error	< 0.1
Predictive stability	PSIS and time-block sensitivity	Exact refits complete; preference > 2 SE and stable

5 Validated results

5.1 Software and artifact validation

The canonical smoke bundle completed in 12.872 s and passed its artifact acceptance with no missing or invalid required files. It records EvoXRB 0.2.0, Python 3.14.2, Windows 11, a 236-by-600 response representation, and 190 fitted detector channels. The environment ledger includes NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.3, Matplotlib 3.10.9, emcee 3.1.6, and dynesty 3.1.0. At manuscript generation, the standard validation suite returned 150 passed; 2 skipped. The focused full-acceptance suite returned 1 passed in 57.29 s and therefore satisfied its preregistered five-seed and parameter-error thresholds.

5.2 Optimization and parameter recovery

Table 6 and Figure 5 show the key numerical result. The deliberately truncated raw genetic search is not a reliable endpoint: both 512 s correct-model realizations exceeded $\Delta C = 1$. Local polishing moved both into numerical agreement with the multi-start reference. The local reference itself required fewer evaluations and less wall time in this low-dimensional educational problem; the scientific value of the genetic stage is therefore basin exploration and inspectability, not a universal speed claim.

Table 6: Synthetic recovery summary. RMSE columns retain native parameter units. The 4096 s row contains one realization and is descriptive only.

Scenario	Stage	n	Fail	Mean ΔC	Time (s)	RMSE: T_{in} ; N_{disk} ; Γ ; K
Correct, 512 s	GA raw	2	1.00	2839.36	0.329	0.4371; 5685.1; 0.2824; 0.1646
Correct, 512 s	GA + L-BFGS-B	2	0.00	1.27×10^{-8}	0.391	0.00206; 283.29; 0.01290; 0.00582
Correct, 512 s	multi-start L-BFGS-B	2	0.00	0	0.235	0.00206; 283.29; 0.01290; 0.00582
Cutoff truth / power-law fit, 512 s	GA + L-BFGS-B	2	0.00	1.21×10^{-9}	0.379	0.06019; 921.57; 0.13737; 0.01743
Correct, 4096 s	GA + L-BFGS-B	1	0.00	1.15×10^{-7}	0.389	0.00161; 95.35; 0.00115; 0.00097

The misspecified fits are the sharper scientific control. After polishing, their mean ΔC from the best value within the wrong model was approximately 10^{-9} , yet the mean biases were +0.0580 keV in T_{in} , -734 in N_{disk} , +0.1346 in Γ , and -0.0150 in K . A numerically exact optimizer can therefore produce physically biased estimates when the spectral family is wrong. This result motivates the preregistered reflection and absorption variants in v0.3.

Validated synthetic evidence from the canonical smoke bundle

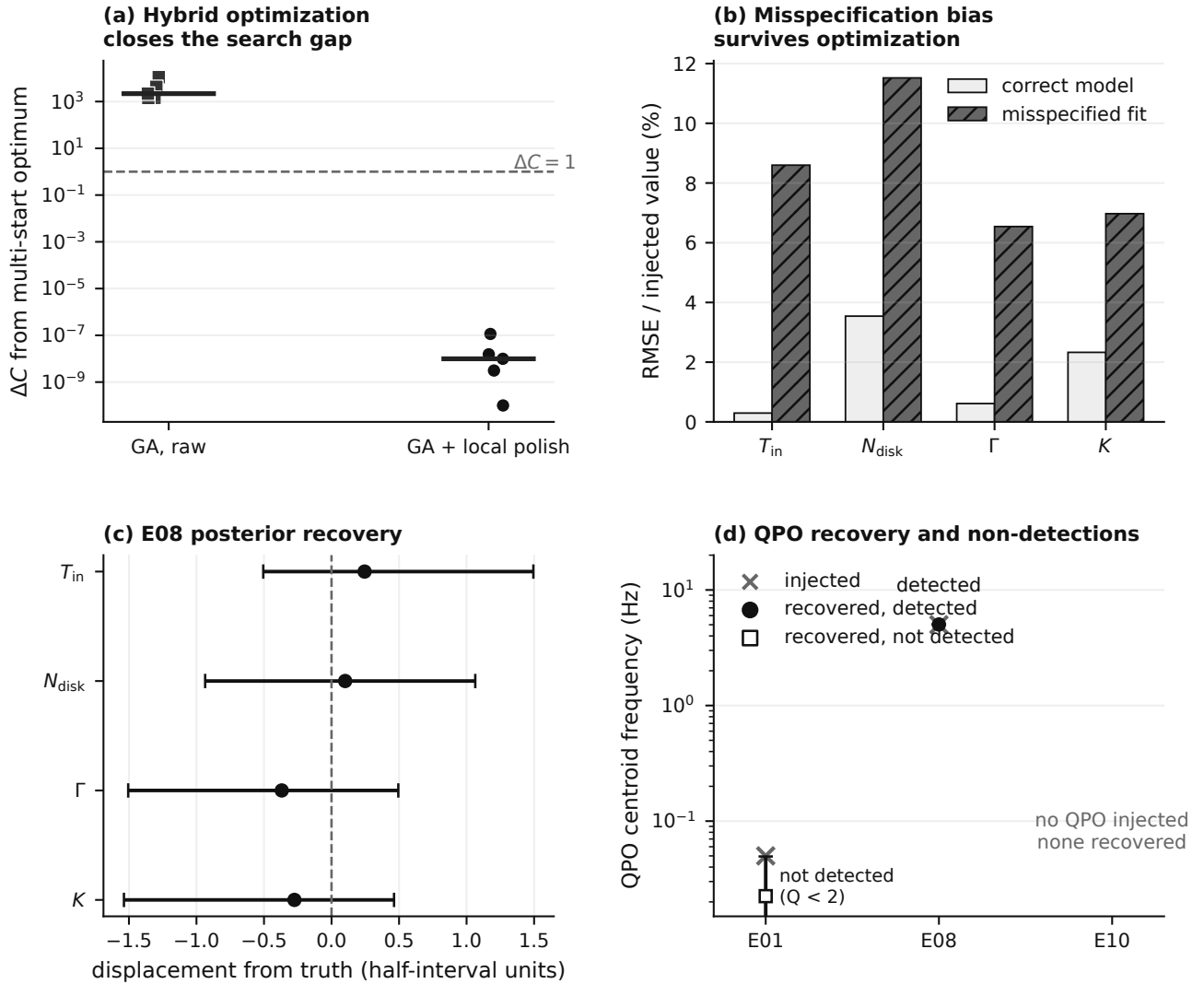


Figure 5: Validated synthetic evidence. (a) The short raw genetic search remains far from the multi-start optimum, while local polishing closes the gap. (b) Correct-model and misspecified-model RMSE as a percentage of the injected value; reaching the fitted-model optimum does not remove misspecification bias. (c) E08 posterior intervals expressed relative to their half-width contain the injected values. (d) The 5 Hz injection is detected, the low- Q 0.05 Hz recovery is retained as a non-detection, and the no-QPO epoch remains empty.

5.3 Posterior recovery

The E08 ensemble posterior completed successfully. Table 7 reports the 16th, 50th, and 84th percentiles; every injected value falls inside the central interval. These intervals characterize one synthetic epoch and should not be read as empirical coverage. Coverage is reserved for the full blinded v0.3 campaign.

Table 7: E08 synthetic posterior recovery.

Parameter	Truth	16th	Median	84th
T_{in} (keV)	0.6500	0.6460	0.6519	0.6618
N_{disk}	8000	7668.6	8035.8	8376.9
Γ	2.1500	2.1272	2.1444	2.1575
K	0.2700	0.2595	0.2681	0.2731

5.4 Timing recovery and non-detections

The 5 Hz E08 QPO was recovered at 5.02770 ± 0.00614 Hz with FWHM 0.60771 Hz, $Q = 8.273$, and amplitude significance 23.58; it passed the synthetic type-C-like gates. E01 returned 0.02242 ± 0.02701 Hz with $Q = 0.606$ and significance 3.12. Despite nominal amplitude significance, it failed the registered quality requirement and remained a non-detection. E10 had neither an injected nor recovered QPO. The result demonstrates why a single significance number cannot replace a component definition and why non-detections must persist into later inference.

Table 8: Synthetic timing validation.

Epoch	Injected (Hz)	Recovered (Hz)	Q	Significance	Detection	Reason
E01	0.05	0.02242 ± 0.02701	0.606	3.12	No	Fails $Q \geq 2$
E08	5.00	5.02770 ± 0.00614	8.273	23.58	Yes	Synthetic type-C-like
E10	none	none	—	0.00	No	No component injected or selected

6 Discussion

6.1 What has been learned across the three generations

First, forward folding and explicit count likelihoods are not implementation details; they define the observable. The synthetic response made this dependence easy to inspect, while the v0.3 OGIP path makes it instrument-correct.

Second, stochastic global optimization and local curvature exploitation serve different purposes. The smoke profile intentionally exposes the failure of an under-budgeted genetic endpoint. The hybrid succeeds because the global stage proposes a basin and the local stage resolves it. In a four-parameter smooth test problem, multi-start L-BFGS-B is faster. The paper therefore makes no claim that genetic algorithms dominate local methods; it claims that the combined workflow is diagnosable and that its best known solution can be independently checked.

Third, optimizer validation is weaker than model validation. The cutoff-truth control reaches the wrong-model optimum while moving several parameters systematically. That observation is the principal bridge from v0.2 to v0.3: a publishable coupling analysis needs physical model variants, calibrated responses, posterior propagation, and predictive checks, not only a lower statistic.

Fourth, timing selection must preserve ambiguity. The E01 example has a nominal amplitude significance above three but a quality factor below the component definition. Treating it as a detection would manufacture a centroid for the population regression. Evidence mixtures and retained non-detections are therefore central, not secondary.

Fifth, implementation completeness and scientific completeness are distinct. The v0.3 analysis path can validate manifests, block missing assets, construct campaigns, fit hypotheses, and enforce gates. It cannot create a trustworthy

XSPEC grid or calibrated mission products in their absence. Reporting a blocked state is a successful safety behavior.

6.2 Relationship to prior J1820 results

The registered alternatives are motivated by the positive $\nu_{\text{QPO}}-\Gamma$ and negative $\nu_{\text{QPO}}-\tau$ relations reported by Li et al. [13]. They are not designed to guarantee the same conclusion. The model adjusts for luminosity and phase, propagates shared distance, compares predictive performance, retains non-detections, and tests contiguous time blocks. Consequently, a null result, a single-descriptor preference, a dual-descriptor preference, or instability across spectral variants is an admissible outcome.

Even if an association is stable, Equation 29 is not a causal model. Photon index and optical depth are model-derived and covariant; both may respond to a changing geometry or accretion rate. Precession, disk truncation, corona contraction, and propagation scenarios can make overlapping qualitative predictions [2, 3, 9]. The protocol therefore uses the language of conditional predictive association.

6.3 Limitations

The validated quantitative results are small synthetic tests. The correct-model 512 s recovery has two realizations; the 4096 s case has one. They demonstrate executable behavior, not frequentist operating characteristics. The educational response omits calibration complexity, the released cutoff surrogate is not ThComp, and the synthetic timing model is not an exhaustive description of J1820 variability.

The v0.3 protocol is also constrained by the quality of external reductions and the fidelity of the physical grid. Reflection models carry systematic uncertainties, fixed spin/inclination choices can couple to inner radius, and shared-free N_{H} does not span every calibration uncertainty. PSIS-LOO diagnostics reduce, but do not eliminate, sensitivity to influential epochs. The full calibration and exact-refit gates are therefore non-waivable.

7 Conclusion

EvoXRB has progressed from a transparent synthetic optimizer to a reproducible observable workflow and then to an implementation-complete, release-gated protocol for testing disk–corona/QPO associations in MAXI J1820+070. The validated v0.2 evidence supports four conclusions:

1. the response-folded Poisson pipeline is deterministic and artifact-complete under its recorded seed and configuration;
2. a budget-limited raw genetic search is insufficient, while local polishing reproduces the multi-start optimum in the tested cases;
3. numerical optimization does not protect against physical model misspecification; and
4. timing logic correctly distinguishes a high-quality 5 Hz synthetic QPO from a low- Q candidate and a no-QPO epoch.

The proposed v0.3 study converts those lessons into a falsifiable observational design: immutable OGIP inputs, a pinned and independently validated XSPEC grid, spectral/timing posterior propagation, four registered hypotheses, PSIS-LOO plus time-block validation, prior/model sensitivity, blinded calibration, and explicit release gates. Until those external requirements pass, the scientifically correct result is “not yet evaluated.” No observational disk–corona/QPO coupling claim is made here.

Software and data availability

The manuscript source, reference database, figure sources, canonical synthetic results, configuration, and validation records are archived with the project. The canonical synthetic evidence is identified by master seed 1,820,070, profile `smoke`, and configuration hash `d17357d31d720259`. The real MAXI/GSC spectrum is a context-only reference and is never used in the synthetic likelihood. External HXMT/NICER products and the physical-grid validation pack were unavailable for the analyses reported here. Documentation and reproducibility guidance are maintained in the README linked after the references.

Acknowledgements

This research has made use of MAXI data provided by RIKEN, JAXA and the MAXI team. The software uses the scientific Python ecosystem, including SciPy [36], and follows community-oriented open-source practices exemplified by Astropy [43].

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Project repository

<https://github.com/svsam/EvoXRB>